

Some leading manufacturers of sensors for aviation

- Active Sensors
- AMETEK Inc.
- BEI Sensors
- Bharat Electronics Ltd
- Crane Aerospace & Electronics
- Esterline Corp.
- Honeywell
- Hydra-Electric
- Komoline Aerospace Ltd
- Magnetic Sensors Corp.
- Meggitt SA
- Omega Engineering
- Oxsensis Ltd
- Rockwell Collins Inc.
- Sensata Technologies Inc.
- SST Sensing Ltd
- Thales Group
- Watson Industries

or above the surface of Earth. Avionics may use satellite-based, ground-based or other systems. Navigation systems calculate the position automatically and display it to the flight crew on moving map displays.

Monitoring. Glass cockpits or computer monitors are used in modern aircraft, instead of gauges and other analogue displays. Cockpit equipment include control, monitoring, communication, navigation, weather and anti-collision systems. Honeywell makes advanced monitoring and display systems for aircraft, helicopters and space vehicles. Ametek makes cockpit indicators and display systems.

Auto control systems. Modern aircraft have autopilot mode to automatically control flight. Most commercial planes are also equipped with aircraft flight control systems to reduce pilot errors and workload at landing or takeoff. In helicopters, auto-stabilisation is used in a similar fashion. The advent of electromechanical systems has increased safety.

Automatic traffic control. Automatic traffic control ensures sufficient space between two or more aircraft, either horizontally or vertically, to prevent collisions. Controllers may coordinate position reports provided by pilots. Radars are also used to check positions of aircraft. Central and control towers, oceanic controllers and terminal controllers enable automatic traffic control.



Fig. 6: Example of portable collision avoidance system (Credit: <https://en.wikipedia.org>)



Fig. 7: A typical black box in aircraft (Credit: <https://commons.wikimedia.org>)



Fig. 8: Health and usage monitoring system for helicopters (Credit: <https://meggittsensing.com>)

Collision avoidance systems. To supplement air traffic control, most aircraft use traffic alert and collision avoidance systems (TCAS) to detect nearby aircraft and prevent mid-air collisions. On terrain, ground-proximity warning systems or radar altimeters are usually employed. Some leading manufacturers of such systems are Rockwell Collins, Honeywell and Thales Group.

A less accurate but inexpensive device is the portable collision avoidance system (PCAS). Pilots all over the world use it. It is a passive device similar in function to a traffic alert and

collision avoidance system.

Flight recorders. Commercial aircraft cockpit data recorders, commonly known as black boxes, store flight information and audio from the cockpit. These are often recovered from an aircraft after a crash to determine control settings and other parameters during the incident. Komoline is an Indian firm that manufactures black box recorders.

Weather systems. Weather systems such as weather radar and lightning detectors or meteorological instruments are used by pilots to view the weather ahead. Incidences like heavy precipitation or severe turbulence due to lightning activity sensed by radars allow pilots to deviate flight paths.

Aircraft management systems. These provide centralised control of the multiple complex systems fitted in aircraft, including engine monitoring and management systems. Health and usage monitoring systems (HUMS) are integrated with aircraft management computers to give maintainers early warnings of parts that need replacement.

Military mission control systems. Military aircraft are designed to deliver weapons or monitor other weapon systems. The vast array of sensors available to the military is used for any and all tactical means. Bigger sensor platforms have mission-management computers. Many electronic sensor components for civil and military applications are available from Bharat Electronics Ltd.

To sum up

Aircraft safety requires feedback on a wide range of flight conditions as well as the states of various flight equipment and systems. Sensors in aircraft play an important role in monitoring the risks associated with aviation activities, operation of aircraft, controlling aircraft to acceptable levels, air traffic control to communicate with aircraft to help maintain separation between two or more aircraft to prevent collision and so on. Hundreds of types of sensors are installed in aircraft to monitor different conditions. These feed information to flight computers and displays for the pilots to handle aircraft effectively. **EFY**